# Fraction Interpretation — Achievers / HOTS

A harder follow-on to `25-fraction-interpretation/`: multi-step word problems (fraction of a fraction, three-category splits), comparing and ordering unlike fractions, adding related-denominator fractions, mixed-number arithmetic, and visual questions that test true conceptual understanding (equivalent fractions shown with different numbers of parts, and the classic "equal parts" misconception trap). 15 questions, 4-option MCQ, untimed.

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**Q1.** In a garden there are 45 flowers. 2/5 are roses, and the rest are lilies. How many lilies are there? A) 27 B) 18 C) 22 D) 20

**Q2.** A fruit basket has 36 fruits. 1/3 are apples, 1/4 are bananas, and the rest are oranges. How many oranges are there? A) 15 B) 21 C) 12 D) 9

**Q3.** Which is greater: 2/3 or 3/5? A) 2/3 B) 3/5 C) They are equal D) Cannot be compared

**Q4.** 1/2 + 1/4 = ? A) 3/4 B) 2/6 C) 1/6 D) 2/4

**Q5.** 1/3 + 1/6 = ? (Give your answer in its simplest form.) A) 1/2 B) 3/6 C) 1/9 D) 2/9

**Q6.** 3/4 of a 40-item collection are red. Of the red items, 1/3 are circles. How many red circles are there? A) 10 B) 30 C) 13 D) 15

**Q7.**



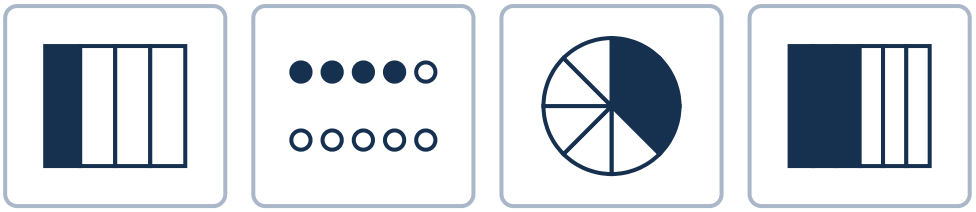
In the group of shapes below, what fraction, in its SIMPLEST FORM, of the shapes are circles?

A) 5/8 B) 10/16 C) 3/8 D) 8/5

**Q8.** Arrange in increasing order: 1/2, 2/3, 1/4 A) 1/4, 1/2, 2/3 B) 1/2, 1/4, 2/3 C) 2/3, 1/2, 1/4 D) 1/4, 2/3, 1/2

**Q9.**

**Which diagram shows exactly 1/2 shaded?**
**agram uses a different number of parts — check the FRACTION, not just the**
**Which option (A–D) is correct?**



A B C D

**Q10.** A ribbon 24 m long is cut so that 5/8 of it is used for decoration. How many metres are LEFT (not used)? A) 9 B) 15 C) 19 D) 12

**Q11.** 2 1/2 + 1 1/2 = ? A) 4 B) 3 1/2 C) 3 D) 4 1/2

**Q12.** A recipe needs 11/4 cups of flour. Express this as a mixed number. A) 2 3/4 B) 1 3/4 C) 2 1/4 D) 3 3/4

**Q13.** There are 40 marbles. 2/5 are blue, 1/8 are red, and the rest are green. How many green marbles are there? A) 19 B) 21 C) 16 D) 24

**Q14.** A group of shapes has 8 shapes, 5 of which are circles (so 5/8 of the group are circles). If 2 more circles are added to the group (now 10 shapes in total), what fraction of the group is now circles, in simplest form? A) 7/10 B) 5/8 C) 7/8 D) 9/10

**Q15.**

**Which diagram correctly shows 5/6 shaded?**
**arefully at both the NUMBER of parts shaded and whether the parts are equa**
**Which option (A–D) is correct?**



A B C D

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*Answer key:* `answer-key.md`

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